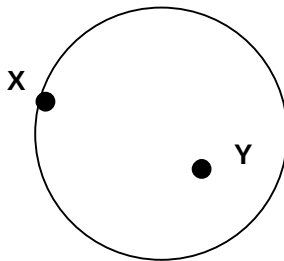


- 11 Sand is sprinkled on a turntable on points X and Y. The turntable is rotating with increasing speed.



Which one of the following comparing sand at points X and Y is true?

	angular speed	linear speed	which will fly off first?
A	$X = Y$	$X > Y$	X
B	$X = Y$	$X > Y$	Y
C	$X > Y$	$X = Y$	Y
D	$X > Y$	$X > Y$	Y

Ans: A

ω on the same turntable is the same for both points.

$v = r\omega$, since $r_x > r_y$, therefore $v_x > v_y$

Centripetal force $F = mr\omega^2$ required for Sand at X is higher, so the sand at X will have a tendency to fly off 1st as the frictional force is unable to provide the higher centripetal force.

- 12 A model car moves in a circular path of radius 0.8 m at an angular speed of $0.5\pi \text{ rad s}^{-1}$